

DOCTORS NETWORK

Master Product, Technical Architecture & Development Blueprint

Version 2.0 | Development-Ready Single Source of Truth | MVP Foundation

Canonical Product Formula

Doctor Professional Network + Doctor Career Marketplace + Doctor Availability Exchange

Surface	Technology Decision	Primary Users
Doctor Mobile App	Android Native - Java + XML; Retrofit/OkHttp; Room; Glide; FCM	Doctors
Hospital CRM	React + TypeScript + Vite	Hospital Admin / HR
Operations Admin	Django Admin for MVP; custom React operations UI later only if justified	Platform operations
Backend	Python + Django + DRF modular monolith	All clients via /api/v1
Data/Async	PostgreSQL + Redis + Celery + S3	Transactional data, queue/cache, files

Document Authority

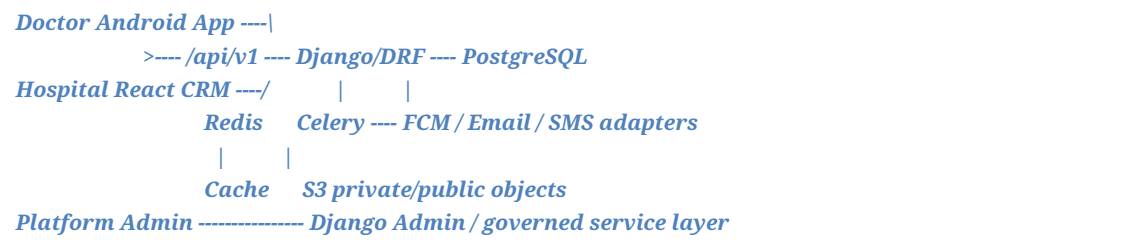
This document supersedes the three source documents for implementation. Source facts are preserved; duplicate requirements are consolidated; architecture recommendations are labelled; unresolved business choices are marked Decision Required.

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1. Product & System Overview

Doctors Network is a doctor-exclusive professional identity, networking, recruitment and workforce availability platform for India. It is explicitly not a patient app, appointment platform, telemedicine product, hospital ERP or generic social network.

Actor / Surface	Responsibility
Doctor Mobile App	Verified professional identity; doctor network/feed/communities; jobs/applications; availability/shift requests; messaging/notifications/privacy.
Hospital CRM	Verified employer profile; jobs; candidates; hiring pipeline; candidate discovery; urgent workforce requirements; shift confirmation; messages/team/settings.
Platform Admin	Manual doctor/hospital verification, moderation, reports, communities, account restrictions, support, operational oversight.
Backend	Authentication, authorization, validation, business workflows/state machines, matching, privacy enforcement, notifications/events, audit, persistence.
Database	Transactional source of truth. PostgreSQL owns canonical state; Redis/FCM/analytics are never canonical.



Architecture Decision
Use a modular monolith for MVP. Keep explicit domain boundaries and service interfaces. Do not start with microservices or Kubernetes.

2. Scope, Assumptions & Decisions

MVP In Scope

- Doctor OTP registration, manual verification and professional profile
- Doctor search, connections, professional feed, posts/comments/save/share/report
- Admin-created specialty communities
- Hospital registration, verification, profile, team and recruitment dashboard
- Jobs, one-tap apply, application tracking and candidate pipeline
- Deterministic doctor-job matching
- Doctor availability calendar, urgent hospital requirements, shift requests
- Professional messaging, persistent in-app notifications + FCM push
- Privacy controls, block/report, audit and operational admin

Out of MVP

- Patients, appointments, telemedicine, prescriptions, diagnosis
- Pharmacy, payroll, attendance, billing and hospital ERP
- Video/audio calling and disappearing messages

- Nurses/students as users
- Courses/education platform
- Generative AI dependency, blockchain, Kubernetes, international recruitment

Assumption A1

Doctor Android application is the only native mobile client in MVP. Hospital users use the React web CRM.

Assumption A2

OTP is the primary doctor authentication mechanism. Hospital users may use OTP in MVP, but stronger authentication/2FA is recommended before broad production rollout.

Decision Required D1

Hospital subscription payments are described commercially, but no payment gateway, invoice/tax workflow, auto-renewal or refund rules are specified. Build Plan/Entitlement/Subscription data model now; do not implement online payment processing until commercial rules and gateway are approved.

Decision Required D2

Verification document upload is not consistently required by the source docs. MVP source flow can be manual register lookup using registration number/council/year. If documents are required operationally, define document types, retention and reviewer policy before enabling upload.

3. Roles & Permissions

Role	Mobile	CRM	Create/Edit	Approve/Reject	Sensitive Access
DOCTOR	Yes	No	Own profile/posts/availability; applications; messages	Accept/decline connections and shifts	Own private career/compensation ; explicitly shared data
HOSPITAL_ADMIN	No	Yes	Hospital profile, jobs, requirements, team	Candidate pipeline; confirm shift	Permitted hospital candidate data
HOSPITAL_HR	No	Yes	Jobs/candidates/requirements per assigned scope	Pipeline actions per permission	Permitted candidate data only
VERIFICATION_ADMIN	No	Admin	Verification records	Doctor/hospital verify/reject	Verification evidence only
MODERATION_ADMIN	No	Admin	Moderation actions	Restrict content/accounts per policy	Reported content/context
SUPPORT_ADMIN	No	Admin	Support actions only	No verification unless granted	Minimum necessary data
SUPER_ADMIN	No	Admin	Platform configuration	All governed operations	Audited privileged access

- Backend performs object-level authorization on every resource. UI hiding is not authorization.
- Hospital users are tenant-scoped to hospital/branch assignments.
- Phone, personal email, expected compensation and detailed availability are never returned by public profile endpoints.
- Admin actions that change verification, moderation, application/shift state or access must be audited.

4. Architecture & Module Boundaries

*apps/
accounts/ doctors/ hospitals/ network/ posts/ communities/
jobs/ applications/ matching/ availability/ workforce/
messaging/ notifications/ moderation/ subscriptions/ audit/ core/*

- HTTP View -> Serializer/Validator -> Domain Service -> Django ORM -> PostgreSQL.
- Use database transactions for apply, status transitions, shift confirmation and subscription changes.
- Side effects publish domain events after commit; Celery delivers push/email, media processing and analytics.
- Search, matching, media and notification logic live behind services to permit later implementation replacement.

5. Complete Module Breakdown

5.1 Authentication & Account Lifecycle

Create secure sessions and bind a user to a doctor or hospital-user identity.

Area	Specification
Actors	Doctor, Hospital User, Admin
Features	Send OTP; verify OTP; issue/refresh/revoke JWT; device registration; logout; account state.
Business / Permission Rules	OTP expiry + attempt limits; one active account identity per verified phone policy; disabled users cannot refresh; roles are server-owned.
Dependencies	User, OTPChallenge, RefreshSession, DeviceToken.
Edge Cases / Errors	Expired/wrong OTP; resend throttling; duplicate phone/email; revoked refresh token; disabled account.

5.2 Doctor Identity & Verification

Create one verified professional identity per doctor.

Area	Specification
Actors	Doctor, Verification Admin
Features	Minimal registration; profile; registration; qualifications; specialization; experience; career preferences; verification submission/status.
Business / Permission Rules	Verification: UNVERIFIED -> PENDING -> VERIFIED/REJECTED; registration number/council/year required; sensitive registration display policy-controlled.
Dependencies	Auth, masters, files if evidence enabled, audit.
Edge Cases / Errors	Duplicate registration; rejected resubmission; incomplete profile; conflicting registration data.

5.3 Professional Network

Doctor-to-doctor professional relationship graph.

Area	Specification
Actors	Doctor
Features	Search/view doctor; connect; accept/ignore; remove; block.
Business / Permission Rules	Unique unordered doctor pair; blocked relationship prevents connection/message interactions.
Dependencies	Doctor identity, privacy, moderation.
Edge Cases / Errors	Self-connect; duplicate pending request; request after block; hidden profile.

5.4 Feed & Clinical Discussion

Professional content with patient-privacy guardrails.

Area	Specification
Actors	Doctor, Moderator
Features	Create text/image/document post; controlled category; like/comment/save/share/report.
Business / Permission Rules	Single Like reaction in MVP; clinical-case posts require privacy confirmation; no video/polls/live.
Dependencies	Identity, media, moderation, notifications.
Edge Cases / Errors	Empty post; invalid category/media; blocked author; removed content; privacy confirmation missing.

5.5 Specialty Communities

Admin-governed specialty spaces.

Area	Specification
Actors	Doctor, Admin
Features	Browse; join/leave; community feed; admin create/edit/archive.
Business / Permission Rules	Doctors cannot create public communities in MVP.
Dependencies	Identity, posts, moderation.
Edge Cases / Errors	Archived community; duplicate membership; unauthorized moderation.

5.6 Hospital Organization & Verification

Represent verified employers and recruitment users.

Area	Specification
Actors	Hospital Admin/HR, Verification Admin
Features	Register hospital; profile; departments; team; verification; public profile.
Business / Permission Rules	Hospital tenant owns its jobs/candidates/requirements; verified badge only after approval.
Dependencies	Auth, audit, files if evidence enabled.
Edge Cases / Errors	Duplicate organization; unverified hospital attempts restricted operation; team invite collision.

5.7 Jobs Marketplace

Structured doctor-specific vacancies.

Area	Specification
Actors	Doctor, Hospital Admin/HR
Features	Create/edit/publish/close jobs; browse/search/save; recommended/nearby/urgent/applied views.
Business / Permission Rules	Only authorized hospital users mutate owned jobs; closed/expired jobs cannot accept applications.
Dependencies	Hospital, masters, matching.
Edge Cases / Errors	Invalid experience/salary ranges; positions <=0; publish incomplete job; expired job.

5.8 Applications & Recruitment CRM

Fast application and transparent hiring pipeline.

Area	Specification
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Actors	Doctor, Hospital Admin/HR
Features	One-tap apply; optional CV; view timeline; candidate list/detail; shortlist/interview/offer/hire/reject/withdraw.
Business / Permission Rules	One application per doctor+job; allowed state transitions only; history immutable.
Dependencies	Jobs, doctor profile, notifications, messaging.
Edge Cases / Errors	Duplicate apply; withdrawn then reapply policy; invalid transition; job closes after application.

5.9 Smart Match Engine

Explainable deterministic match score for jobs and candidates.

Area	Specification
Actors	Doctor, Hospital Admin/HR
Features	Compute factor score and recommendations.
Business / Permission Rules	MVP weights: qualification 25, specialization 25, experience 15, location 15, availability 10, job preference 10. Missing data must not be fabricated.
Dependencies	Doctor profile, job, availability.
Edge Cases / Errors	Insufficient profile data; stale score; location not available; no matches.

5.10 Availability Exchange

Structured doctor availability without live tracking.

Area	Specification
Actors	Doctor, verified Hospital users
Features	Set availability type; slots; preferred location/radius; work type; private minimum compensation; hospital search.
Business / Permission Rules	Detailed availability primarily visible to verified institutions; compensation private; professional location/radius only, no exact live GPS.
Dependencies	Doctor profile, privacy, matching.
Edge Cases / Errors	Overlapping slots; end before start; unavailable doctor returned from stale cache.

5.11 Urgent Requirement & Shift Lifecycle

Fill temporary/urgent coverage.

Area	Specification
Actors	Doctor, verified Hospital users
Features	Create requirement; find matches; send request; doctor accept/decline; hospital confirm; complete/cancel.
Business / Permission Rules	REQUESTED -> ACCEPTED_BY_DOCTOR -> CONFIRMED_BY_HOSPITAL -> COMPLETED; CANCELLED terminal branch.
Dependencies	Availability, matching, notifications, messaging, audit.
Edge Cases / Errors	Slot conflict after acceptance; duplicate request; cancellation after confirmation; requirement filled.

5.12 Messaging

Professional conversations linked to doctor/hospital workflows.

Area	Specification
Actors	Doctor, Hospital users
Features	Conversation list/thread; text/image/document; unread count.
Business / Permission Rules	Only participants can read/send; block policy applies; no

	audio/video/disappearing messages.
Dependencies	Identity, media, notifications.
Edge Cases / Errors	Nonparticipant access; removed attachment; blocked participant; message send retry.

5.13 Notifications

Persistent event inbox plus delivery channels.

Area	Specification
Actors	Doctor, Hospital users
Features	In-app list/read; push; preferences; deep links.
Business / Permission Rules	Notification DB row is source of truth; FCM is best-effort delivery.
Dependencies	All event-producing modules.
Edge Cases / Errors	Invalid device token; duplicate delivery; stale deep link; user disabled category.

5.14 Search & Discovery

Discover doctors, hospitals, jobs and communities.

Area	Specification
Actors	Doctor; Hospital users for candidate search
Features	Global search + filters; candidate discovery.
Business / Permission Rules	Expected compensation never public filter; hospital candidate visibility respects doctor consent/privacy.
Dependencies	Identity, jobs, communities, privacy.
Edge Cases / Errors	No results; invalid filter; deep pagination; restricted profile.

5.15 Privacy, Consent, Block & Report

Enforce visibility, consent and trust.

Area	Specification
Actors	Doctor, Hospital users, Admin
Features	Visibility settings; selected hospitals; block/report; data export/delete request.
Business / Permission Rules	Profile: Everyone/Doctors Only/Connections Only; career: Verified Hospitals/Selected/Hidden; sensitive data minimized.
Dependencies	All domains, audit.
Edge Cases / Errors	Deletion with legal/audit retention; blocked users in existing conversations; consent withdrawal.

5.16 Subscription & Entitlements

Represent hospital plan capabilities without hardcoding.

Area	Specification
Actors	Hospital Admin, Super Admin
Features	Plan, features, subscription status, entitlement checks.
Business / Permission Rules	Doctors free in MVP; plan capability checked by backend. Payment processing deferred until D1 resolved.
Dependencies	Hospital, audit.
Edge Cases / Errors	Expired plan; downgrade over quota; manual admin grant.

6. Mobile App Frontend Specification

Bottom navigation: Home | Network | Jobs | Availability | Profile. Android architecture: Java/XML + repository/use-case boundaries; Activities/Fragments do not call Retrofit directly.

Screen	Purpose	Navigation	UI / Required Data	API Mapping	States / Actions	Next
Splash / Session Restore	Resolve token/account/onboarding state	App launch	Logo/progress; no form	GET /auth/session or local token + refresh	Loading spinner; offline fallback to cached session; auth failure -> Login	Login / onboarding / Home
Mobile Number	Start OTP	Splash/logout	Country code + mobile	POST /auth/send-otp/	Inline validation; throttled resend error	OTP Verification
OTP Verification	Verify ownership	Mobile Number	6-digit OTP, resend timer	POST /auth/verify-otp/	Invalid/expired OTP; loading; success token persistence	Account Type / Home
Doctor Registration	Minimal professional identity	First login	Name + registration + council/year + email	POST /doctors/onboarding/	Field errors; duplicate registration conflict	Verification/ Profile completion
Home Feed	Professional feed	Bottom Home	Feed, create post, interactions	GET /feed/	Skeleton; empty feed; retry	Post detail / Create Post
Create Post	Publish professional content	Home/ community	Category, content, clinical-case flag, media	POST /posts/	Upload progress; privacy confirmation error	Post Detail
Network	Discover/manage connections	Bottom Network	Search, suggestions, requests	GET /doctors;/connections/	Empty results/requests	Doctor Profile
Jobs	Discover jobs	Bottom Jobs	Tabs Recommended/Nearby/Urgent/Saved/Applied	GET /jobs/	Empty tab; retry	Job Detail
Job Detail	Evaluate/apply	Jobs/notification	Job, hospital, match, save/apply	GET /jobs/{id}; POST /jobs/{id}/apply/	Closed/already applied	Application Detail
Applications	Track hiring	Jobs Applied	Application list + timeline	GET /applications/me/	Empty applications	Application Detail
Availability	Publish work availability	Bottom Availability	Status, calendar, slots, radius, type, compensation	GET/PUT /availability/	Slot validation/conflict	Shift Requests
Shift Request Detail	Respond to opportunity	Notification/ Availability	Hospital, role, date/time, location, compensation	POST accept/decline	Already filled/cancelled/conflict	Availability / Message
Messages	Professional communication	Notifications/ profile/job/shift	Conversation list/thread/composer	/ conversations/ /messages/	Offline/retry/ upload error	Thread
Notifications	Persistent event inbox	App icon/deep link	Notification list/read	GET/PATCH /notifications/	Empty state	Deep-linked screen
Profile & Settings	Manage identity/privacy	Bottom Profile	Profile sections, verification, privacy, account	/doctors/me/ + privacy APIs	Partial profile, save error	Section editor

Doctor Registration - exact field contract

Field / API key	Type	Req?	Validation / Default	Editable / Visibility
first_name	string	Required	trim; 2-80 chars	Editable
last_name	string	Required	trim; 1-80 chars	Editable
phone	E.164 string	Required	Verified by OTP; default	Read-only after OTP

			from auth	unless dedicated change flow
email	email	Required per source registration form	Normalize lowercase; valid email	Editable; private
registration_number	string	Required	trim; uniqueness policy by council	Editable until verification submitted
state_medical_council_id	UUID	Required	Must exist/active in master	Editable until verification submitted
registration_year	integer	Required	reasonable year <= current year	Editable until verification submitted

Doctor Profile - field contract

Field / API key	Type	Req?	Validation / Default	Editable / Visibility
profile_photo	file ref	Optional	image policy	Editable
professional_headline	string	Optional	max 160	Editable/public per profile visibility
about	text	Optional	recommended max 2000	Editable
primary_specialization_id	UUID	Recommended required for match	active specialty	Editable
clinical_interest_ids	UUID[]	Optional	active values	Editable
languages	string[]	Optional	controlled/free list decision	Editable
current_hospital_name	string	Optional	max 200	Editable
professional_city/location	location object	Recommended	no live GPS required	Editable
open_to_opportunities	boolean	Optional	default false	Editable/private gate
career_preferences	relation[]	Optional	Full-time/Locum/Visiting etc.	Editable/private
expected_compensation	decimal + currency	Optional	>=0	Private; only hiring workflow with consent

Availability - exact field contract

Field / API key	Type	Req?	Validation / Default	Editable / Visibility
availability_type	enum	Required	LOCUM/VISITING/TEMPORARY/PART_TIME	Editable
available_from	date	Required	<= available_until	Editable
available_until	date	Required	>= available_from	Editable
slots	array	Required for specific times	start < end; no overlap	Editable
preferred_location	location	Optional	professional work location	Editable/private
preferred_radius_km	integer	Optional	1-500 Assumption	Editable/private
minimum_compensation	decimal	Optional	>=0	Private
currency	string	If compensation	default INR	Editable/private

7. Hospital CRM Frontend Specification

Page	Purpose	Navigation	UI Sections	API	States	Next
Hospital Login	Authenticate HR/admin	Public	Phone/email auth form	Auth APIs	Errors/throttle	Onboarding/Dashboard
Hospital Onboarding	Create organization	First login	Name/type/city/contact/designation/official email	POST /hospital/onboarding/	Duplicate/validation	Verification
Dashboard	Operational summary	Sidebar	Active jobs, new applicants, shortlisted, interviews, urgent requirements	GET /hospital/dashboard/	Skeleton/empty	Jobs/Candidates/Requirements
Hospital Profile	Employer brand	Sidebar	Logo, name, type,	GET/PATCH	Save validation	Public preview

			location, about, beds, departments, facilities	/hospital/profile/		
Jobs List	Manage vacancies	Sidebar Jobs	Status/filter/table/actions	GET /hospital/jobs/	Empty list	Create/Edit/Detail
Create/Edit Job	Publish vacancy	Jobs	Structured job form	POST/PATCH /hospital/jobs/	Draft/save/publish errors	Job Detail
Candidates	Recruitment pipeline	Sidebar/job detail	Filters + candidate cards/table	GET /hospital/candidates/	No candidates	Candidate Detail
Candidate Detail	Evaluate doctor	Candidates	Profile, match, application history, actions	GET /hospital/applications/{id}/	Restricted private fields	Pipeline action/Message
Availability Exchange	Find available doctors	Sidebar	Filters, matches, requirement CTA	GET /hospital/available-doctors/	No matches	Doctor/Requirement
Urgent Requirement	Create shift need	Availability	Requirement form	POST /hospital/requirements/	Validation	Matches
Shift Requests	Manage requests	Availability	Status list/detail/confirm/cancel	Shift APIs	Already changed conflict	Message/Complete
Messages	Communicate	Sidebar/candidate	Threads	Messaging APIs	Retry	Thread
Team	Manage HR access	Sidebar	Users, roles, branch scope	Hospital user APIs	Permission error	Invite/Edit
Settings	Preferences/security/plan	Sidebar	Notification, security, subscription info	Settings/subscription APIs	Save errors	-

Hospital Onboarding - exact field contract

Field / API key	Type	Req?	Validation / Default	Editable / Visibility
hospital_name	string	Required	2-200	Editable until verification lock policy
hospital_type	enum	Required	HOSPITAL/CLINIC/NURSING_HOME; other types future	Editable
city/location	location	Required	valid location	Editable
contact_person_name	string	Required	2-120	Editable/private
designation	string	Required	2-120	Editable/private
official_email	email	Required	valid business email; verification optional Decision Required	Editable/private

Create Job - exact field contract

Field / API key	Type	Req?	Validation / Default	Editable / Visibility
title	string	Required	3-160	Editable draft
specialization_id	UUID	Required	active master	Editable
qualification_ids	UUID[]	Required	>=1	Editable
experience_min_years	decimal	Required	>=0	Editable
experience_max_years	decimal	Optional	>= min	Editable
location	location	Required	city/address structure	Editable
salary_min	decimal	Optional	>=0	Editable/private/public by visibility
salary_max	decimal	Optional	>= min	Editable
salary_visibility	enum	Required	PUBLIC/ON_REQUEST/HIDDEN	Editable
job_type	enum	Required	FULL_TIME/PART_TIME/VISITING/LOCUM/CONTRACT	Editable
department_id	UUID	Optional	belongs to hospital	Editable

shift_type	enum	Required	DAY/NIGHT/ ROTATIONAL/FLEXIBLE	Editable
joining_requirement	string/enum	Required	e.g. immediate/date	Editable
positions	integer	Required	>=1	Editable
description	text	Required	max policy	Editable
required_skills	string[]	Optional	normalized	Editable

Urgent Requirement - exact field contract

Field / API key	Type	Req?	Validation / Default	Editable / Visibility
specialization_id	UUID	Required	active specialty	Editable
qualification_ids	UUID[]	Required	>=1	Editable
date	date	Required	not in past at publish	Editable
start_time	time	Required	start/end valid incl overnight handling	Editable
end_time	time	Required	valid duration	Editable
location	location	Required	hospital/provided location	Editable
compensation	decimal	Required	>=0	Editable; visible to requested doctor
currency	string	Required	default INR	Editable
doctors_required	integer	Required	>=1	Editable
urgency	enum	Required	NORMAL/URGENT/ IMMEDIATE	Editable
notes	text	Optional	max policy	Editable

8. Backend/API Specification

All endpoints use /api/v1, JSON except multipart/presigned upload workflows, Bearer JWT where authenticated, UUID public IDs, UTC timestamps in ISO-8601, and consistent pagination/error contracts.

Module	API	Method	Endpoint	Auth/Role	Request	DB Ops	Side Effects / Audit	Priority
Auth	Send OTP	POST	/auth/send-otp/	Public	phone, purpose	OTPChallenge	SMS adapter; audit security event	P0
Auth	Verify OTP	POST	/auth/verify-otp/	Public	phone, otp, device	User/session	issue tokens; register device	P0
Auth	Refresh	POST	/auth/refresh/	Refresh token	refresh_token	RefreshSession	rotate/ revoke token	P0
Doctor	Onboarding	POST	/doctors/onboarding/	DOCTOR	registration fields	DoctorProfile/Registration	audit	P0
Doctor	My Profile	GET/PATCH	/doctors/me/	DOCTOR	profile fields	Doctor tables	profile updated event	P0
Doctor	Verification Submit	POST	/doctors/verification/submit/	DOCTOR	registration/ evidence refs	Verification	notify admin optional	P0
Doctors	Search	GET	/doctors/	Authenticated	q, specialty, location, verified	Doctor indexes	none	P1
Network	Connect	POST	/connections/request/	DOCTOR	receiver_id	Connection	notify receiver	P1
Network	Connection Action	POST	/connections/{id}/{action}/	DOCTOR	action	Connection	notify acceptance	P1
Posts	Feed	GET	/feed/	DOCTOR	cursor, scope	Post indexes	none	P1
Posts	Create	POST	/posts/	DOCTOR	category, content	Post/	followers/	P1

					tent,clinical flag,media	PostMedia	community notification policy	
Posts	Engage	POST	/posts/{id}/ like save/	DOCTOR	-	Like/ SavedPost	optional notify	P2
Comments	Create	POST	/posts/{id}/ comments/	DOCTOR	body	Comment	notify author	P1
Communitie s	List/Join	GET/POST	/ communities /; /communitie s/{id}/join/	DOCTOR	filters	Community/ Membership	none	P1
Jobs	Browse	GET	/jobs/	DOCTOR	tabs/filters	Job	none	P1
Jobs	Detail	GET	/jobs/{id}/	DOCTOR	-	Job + match	none	P1
Jobs	Apply	POST	/jobs/{id}/ apply/	DOCTOR	optional cv_file_id	Application/ Consent	notify hospital	P1
Applications	Mine	GET	/ applications/ me/	DOCTOR	status	Application/ history	none	P1
Hospital	Profile	GET/PATCH	/hospital/ profile/	HOSPITAL_*	profile fields	Hospital tables	audit on verification- sensitive fields	P0
Hospital	Jobs CRUD	GET/POST/ PATCH	/hospital/ jobs/	HOSPITAL_*	job fields	Job	job published event	P1
Hospital	Candidates	GET	/hospital/ jobs/{id}/ candidates/	HOSPITAL_*	filters	Application/ Doctor	none	P1
Applications	Transition	POST	/hospital/ applications/ {id}/ transition/	HOSPITAL_*	to_status,me tadata	Application/ history/ interview/ offer	notify doctor + audit	P1
Availability	Get/Update	GET/PUT	/availability/	DOCTOR	availability+ slots	Availability/ Slots	availability changed event	P1
Workforce	Create Requiremen t	POST	/hospital/ requirements/	HOSPITAL_*	requirement fields	UrgentRequi rement	match job/event	P1
Workforce	Requiremen t Matches	GET	/hospital/ requirements/{id}/ matches/	HOSPITAL_*	filters	read doctor/avail ability	none	P1
Workforce	Send Shift Request	POST	/hospital/ requirements/{id}/ requests/	HOSPITAL_*	doctor_id	ShiftRequest /history	notify doctor + audit	P1
Workforce	Doctor Action	POST	/shift- requests/ {id}/accept decline/	DOCTOR	-	ShiftRequest /history	notify hospital	P1
Workforce	Hospital Confirm	POST	/shift- requests/ {id}/confirm/	HOSPITAL_*	-	ShiftRequest /history	notify doctor + audit	P1
Messaging	Conversatio ns	GET/POST	/ conversations/	Authenticate d	participant/ context	Conversatio n	none	P2
Messaging	Messages	GET/POST	/ conversations/{id}/ messages/	Participant	body/file	Message	notify recipient	P2
Notifications	List/Read	GET/PATCH	/ notifications /	Authenticate d	cursor/read	Notification	none	P2
Files	Create Upload	POST	/files/ uploads/	Authenticate d	purpose,mi me,size	FileAsset	presigned S3	P0
Privacy	Settings	GET/PATCH	/privacy/	DOCTOR	visibility	PrivacySetti	audit	P1

			settings/		fields	ng	consent changes	
Moderation	Report	POST	/reports/	Authenticate	target,reason, details	Report	admin queue	P1
Admin	Verification Decision	POST	/admin/verifications/{id}/decision/	VERIFICATION_ADMIN	decision,reason	Verification/Audit	notify subject	P0

8.1 Standard Request/Response Contract

Authorization: Bearer <access_token>

Content-Type: application/json

X-Request-ID: optional client correlation id

Idempotency-Key: required/recommended for critical creates such as apply/shift request

```
{
  "success": true,
  "data": { ... },
  "meta": {"request_id": "...", "pagination": {...}}
}

{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Please correct the highlighted fields.",
    "fields": {"registration_year":["Must not be in the future."]},
    "request_id": "..."
  }
}
```

8.2 Representative API - Apply to Job

Item	Contract
Method/Endpoint	POST /api/v1/jobs/{job_uuid}/apply/
Auth	DOCTOR; active account
Path	job_uuid UUID
Body	{"cv_file_id": "uuid null"}
Validation	Job published/open/not expired; doctor profile exists; no existing application; CV owned by doctor if supplied
Transaction	Lock/check job as needed -> create JobApplication(APPLIED) -> create ApplicationStatusHistory -> create ConsentRecord for profile sharing
Side effects	Notification/event to hospital; analytics event after commit
Success	201 Created with application UUID, status, created_at
Errors	401 auth; 403 role; 404 job; 409 already applied/job closed; 422 business validation
Audit	Application creation + consent record; no sensitive fields in log

8.3 Representative API - Application Transition

Item	Contract
Method/Endpoint	POST /api/v1/hospital/applications/{uuid}/transition/
Auth	HOSPITAL_ADMIN/HR with hospital ownership
Body	to_status; optional interview datetime/location/notes or offer metadata according to target
Business Logic	Validate transition graph; create dependent Interview/Offer record when required; append immutable status history

DB	Atomic transaction on application + history + optional child entity
Side effects	Notify doctor; audit actor/from/to
Errors	403 tenant; 404; 409 stale/invalid transition; 422 missing target metadata

9. Database Design

All public entities use UUID PKs; created_at/updated_at on mutable tables. The following is the canonical MVP logical schema. Exact varchar lengths may be adjusted during migration review without changing API semantics.

Table	Key Columns / Constraints	Purpose	Indexes
users	id UUID PK; phone varchar unique; email citext nullable/unique policy; user_type enum; status enum; is_staff bool; last_login; created_at; updated_at	Identity root	phone, user_type/status
otp_challenges	id; phone; purpose; otp_hash; expires_at; attempts; resend_count; consumed_at; created_at	OTP security	phone+created_at
refresh_sessions	id; user_id FK; token_hash; device_id; expires_at; revoked_at; created_at	Session revocation	user_id, expires_at
doctor_profiles	id; user_id unique FK; first_name; last_name; photo_file_id; headline; about; primary_specialization_id; professional_location; open_to_opportunities; verification_status	Doctor root	user_id; specialization; verification
doctor_registrations	id; doctor_id FK; council_id; registration_number; registration_year; is_primary; verification_status	Medical registration	unique(council_id,registration_number); doctor_id
doctor_qualifications	id; doctor_id; degree_id/name; institution; year; file_id nullable	Qualifications	doctor_id, degree
doctor_experiences	id; doctor_id; role; hospital_name; start_date; end_date nullable; location; is_current	Experience	doctor_id, dates
doctor_specializations	id; doctor_id; specialization_id; type PRIMARY/INTEREST	Specialty relation	unique doctor+specialization+type
doctor_career_preferences	id; doctor_id; preference_type; preferred_locations jsonb/normalized later; expected_compensation numeric; currency; joining_availability	Private career intent	doctor_id
doctor_privacy_settings	doctor_id PK/FK; profile_visibility; career_visibility; availability_visibility; messaging_permission; connection_permission	Privacy	doctor_id
hospitals	id; name; type; logo_file_id; about; bed_count; location; verification_status; created_at	Organization root	name/location; verification
hospital_branches	id; hospital_id; name; location; is_primary	Branch-ready model	hospital_id
hospital_departments	id; hospital_id/branch_id; name; active	Departments	hospital_id,name
hospital_users	id; user_id unique; hospital_id; branch_id nullable; role; status	Tenant membership	hospital_id,role

connections	id; sender_doctor_id; receiver_doctor_id; pair_key; status; created_at; responded_at	Network	unique pair_key; sender/receiver/status
posts	id; author_doctor_id; community_id nullable; category; body; is_clinical_case; privacy_confirmed_at; status; created_at	Content	author, community, created_at,status
post_media	id; post_id; file_asset_id; sort_order	Post files	post_id
post_likes/comments/saved_posts	normalized engagement tables with unique constraints	Engagement	post/user
communities	id; name; slug unique; specialty_id; status; created_by_admin	Admin communities	slug,status
community_memberships	community_id; doctor_id; joined_at; status	Membership	unique community+doctor
jobs	id; hospital_id; branch_id; title; specialization_id; experience_min/max; location; salary_min/max; salary_visibility; job_type; shift_type; joining_requirement; positions; description; status; published_at; expires_at	Vacancy	hospital,status,specialization,loc ation
job_qualifications/job_skills	job relation tables	Requirements	job_id
job_applications	id; job_id; doctor_id; status; match_score numeric; cv_file_id; applied_at; updated_at	Application	unique(job_id,doctor_id); status
application_status_history	id; application_id; from_status; to_status; actor_user_id; metadata jsonb; created_at	Immutable pipeline history	application_id,created_at
interviews	id; application_id; scheduled_at; mode/location; notes; status	Interview	application_id,scheduled_at
offers	id; application_id; compensation; currency; joining_date; status; metadata	Offer	application_id,status
doctor_availability	id; doctor_id unique; status; availability_type; preferred_location; radius_km; min_compensation; currency; updated_at	Availability header	doctor_id,status
availability_slots	id; availability_id; start_at; end_at; status	Calendar slots	availability_id,start_at,end_at
urgent_requirements	id; hospital_id; specialization_id; date/start/end; location; compensation; currency; doctors_required; urgency; notes; status	Urgent need	hospital,status,date,specialization n
shift_requests	id; requirement_id; doctor_id; status; requested_at; accepted_at; confirmed_at; completed_at; cancelled_at	Shift workflow	requirement,doctor,status
shift_status_history	id; shift_request_id; from_status; to_status; actor; reason; created_at	Shift audit history	shift_request,created_at
conversations	id; context_type/context_id nullable; created_at	Messaging root	context
conversation_participants	conversation_id; user_id; last_read_at; active	Participants	unique conversation+user
messages	id; conversation_id; sender_user_id; body; type; created_at; edited_at nullable; status	Messages	conversation,created_at
file_assets	id; owner_user_id; purpose; storage_key; mime_type; size_bytes; checksum; visibility;	Object metadata	owner,purpose,status

	status; created_at		
notifications	id; recipient_user_id; type; title; body; data jsonb; deep_link; read_at; created_at	Persistent notification	recipient,read_at,created_at
device_tokens	id; user_id; platform; token unique; active; last_seen_at	Push device	user,active
reports	id; reporter_user_id; target_type; target_id; reason; details; status; created_at	Moderation queue	status,target
blocks	id; blocker_user_id; blocked_user_id; created_at	Block graph	unique blocker+blocked
consent_records	id; user_id; purpose; target_type/id; version; granted_at; withdrawn_at	Consent ledger	user,purpose,target
audit_logs	id; actor_user_id nullable; action; entity_type; entity_id; before/after jsonb redacted; ip_hash/context; created_at	Privileged/business audit	entity,actor,created_at
plans/plan_features/hospital_subscriptions	Plan and entitlement data; payment fields deferred	Commercial entitlements	hospital/status/plan

Soft Delete Policy

Do not add deleted_at everywhere. Use lifecycle status for jobs, posts, communities, accounts and subscriptions. Use hard delete only where allowed. Retain immutable audit/history according to approved retention policy.

10. End-to-End Flows

Flow	Sequence
Doctor Verification	Doctor submits registration -> frontend validates -> POST verification -> backend validates ownership/completeness -> transaction creates/updates PENDING verification -> admin queue -> admin decision -> audit -> Notification -> doctor sees VERIFIED/REJECTED.
Job Application	Doctor taps Apply -> app checks local completeness -> POST apply -> backend checks job/open/duplicate/privacy -> atomic application + history + consent -> hospital notification -> 201 -> app updates Applied state/timeline.
Hiring Pipeline	HR selects Shortlist/Interview/Offer/Hire -> CRM validates required metadata -> transition API -> backend tenant + state validation -> transaction status/history/child record -> notification -> CRM refresh -> doctor timeline updates.
Urgent Shift	Hospital creates requirement -> backend stores -> match service returns eligible doctors -> HR sends request -> doctor push/in-app -> accept/decline -> hospital confirm -> completion/cancel transitions -> immutable history/audit.
Clinical Post	Doctor selects Clinical Discussion -> clinical-case flag -> warning + mandatory de-identification confirmation -> media upload -> create post -> backend validates confirmation/category/files -> publish -> moderation/report hooks.

11. Notifications

Type	Recipient	Channel	Backend Event	Deep Link
CONNECTION_REQUEST	Target doctor	In-app + Push	connection.requested	doctor://network/requests
CONNECTION_ACCEPTED	Requester	In-app + Push	connection.accepted	doctor://profile/{doctor_id}
APPLICATION_UPDATE	Doctor	In-app + Push	application.status_change	doctor://applications/{id}
INTERVIEW_UPDATE	Doctor	In-app + Push; email optional	interview.scheduled	doctor://applications/{id}
NEW_MESSAGE	Conversation participant	In-app + Push	message.created	doctor://messages/

				{conversation_id} or CRM route
RECOMMENDED_JOB	Doctor	In-app/Push preference	job.recommended	doctor://jobs/{id}
SHIFT_REQUEST	Doctor	In-app + Push; SMS fallback Decision Required	shift.requested	doctor://shifts/{id}
SHIFT_ACCEPTED	Hospital users	In-app CRM	shift.accepted	/availability/requests/{id}
SHIFT_CONFIRMED	Doctor	In-app + Push	shift.confirmed	doctor://shifts/{id}
VERIFICATION_UPDATE	Doctor/Hospital user	In-app + Push/email optional	verification.decided	relevant verification page

- Create Notification row before/as part of event handling; push delivery result does not determine read state.
- PATCH notification read_at when opened; bulk mark-read endpoint supported.
- Device token invalidation handled asynchronously.

12. File / Image / Document Handling

Purpose	Allowed Types	Recommended Max	Visibility
Profile/Hospital Logo	JPEG, PNG, WebP	5 MB original; generate optimized variants	Public/visibility-controlled
Post Image	JPEG, PNG, WebP	10 MB each; MVP max count Decision Required	According to post visibility
Post Document	PDF	15 MB	According to post visibility
CV	PDF	10 MB	Private; doctor + explicitly shared hospital
Verification Evidence (if enabled)	PDF/JPEG/PNG	10 MB/file	Strict private; verification admins only
Message Attachment	JPEG/PNG/WebP/PDF	10 MB	Conversation participants only

- Recommended upload pattern: client requests upload intent -> backend validates purpose/mime/size -> short-lived S3 presigned upload -> client uploads -> finalize endpoint verifies object metadata/checksum -> FileAsset ACTIVE.
- Never trust file extension or client MIME alone; inspect magic bytes/server metadata.
- Use private S3 objects for CV/verification/message files; issue short-lived signed download URLs after authorization.
- Replace creates a new FileAsset/reference; old unreferenced private object is asynchronously deleted after safe grace period.
- Image compression/thumbnail generation runs asynchronously; original retention policy is configurable.

13. Security Architecture

- JWT access tokens short-lived; rotating refresh sessions stored hashed/revocable.
- OTP values stored hashed; strict expiry, attempt cap, resend cooldown and IP/phone rate limits.
- RBAC + object/tenant authorization on every API; deny by default.
- Server-side schema validation, enum validation, range checks and state-machine validation.
- TLS in transit; AWS-managed encryption at rest for RDS/S3; secrets in secret manager/environment, not repository.
- Sensitive data excluded/redacted from application logs, analytics and audit before/after payloads.
- S3 buckets private by default; signed URLs short-lived; upload keys are server-generated.
- CSRF protection for cookie-auth admin; CRM JWT storage strategy should avoid long-lived tokens in localStorage where possible.
- Hospital HR 2FA is strongly recommended before scale; password policy applies only if password auth is introduced.

- Security headers, CORS allowlist, dependency scanning, backup encryption, least-privilege IAM and admin access logging.
- Abuse controls: rate limit search, messaging, connection requests, posts and reports; anti-enumeration responses for OTP/account lookup.

14. Error Handling

HTTP	Code	Use	Frontend Behavior
400	BAD_REQUEST	Malformed request	Generic retry/correct request
401	AUTH_REQUIRED / TOKEN_EXPIRED	Missing/invalid auth	Attempt refresh once; then login
403	FORBIDDEN	Role/object/privacy denied	Show no-access; never retry blindly
404	NOT_FOUND	Missing or intentionally hidden resource	Show unavailable/not found
409	CONFLICT	Duplicate apply, stale transition, slot conflict	Refresh state; show actionable message
422	VALIDATION_ERROR / BUSINESS_RULE_VIOLATION	Field/business validation	Map fields + banner
429	RATE_LIMITED	OTP/search/write throttled	Show retry_after countdown
500	INTERNAL_ERROR	Unexpected failure	Safe generic message + request_id; retry where idempotent
503	SERVICE_UNAVAILABLE	Dependency outage	Temporary unavailable; queue/retry if safe

15. Development Dependency Mapping

Module	Depends On	Frontend Can Mock?	Must Finalize First
Auth	User/session/device	Yes, after OpenAPI contract	User roles, token contract, OTP rules
Doctor Identity	Auth + master data	Yes	Doctor core schema + verification states
Network	Doctor identity + privacy	Yes	Connection uniqueness/state
Feed	Doctor identity + files + moderation	Yes	Post categories/media policy
Hospital	Auth + tenant model	Yes	Hospital/HospitalUser/branch decision
Jobs	Hospital + masters	Yes	Job schema/status
Applications	Jobs + doctor + consent	Limited	Pipeline state machine
Matching	Doctor/job/availability	No useful final QA until data stable	Scoring rules
Availability	Doctor + privacy	Yes	Slot/time model
Workforce	Hospital + availability + matching	Limited	Shift state machine
Messaging	Identity + block + files	Yes	Conversation participant model
Notifications	Events + devices	Yes	Event/deep-link registry
Subscriptions	Hospital + entitlement	Yes	Plan capability model; payments deferred

16. Exact Development Sequence

Phase	Backend Tasks	Frontend Tasks	Database	Deliverable	Testing Criteria	Definition of Done
Phase 0 - Requirement Freeze & Contracts	Finalize canonical enums/state machines, API naming, masters, Decision	Android/CRM design system + navigation shells using mocks only.	ERD baseline, UUID/time/money/location conventions.	Approved architecture + API contract v0.1.	Review checklist; no unresolved blocker for P0.	All P0 contracts signed off.

	Required register; OpenAPI skeleton.					
Phase 1 - Platform Foundation	Django project, settings, PostgreSQL, Redis/Celery wiring, custom User, base models, error envelope, logging, health endpoints.	Project setup, networking/session abstractions.	Create accounts/core/audit/file foundations.	Runnable dev stack + CI.	Migration + health + error contract tests.	Fresh environment deploys automatically.
Phase 2 - Auth & RBAC	OTP/JWT/refresh/device, rate limits, roles/object permission framework.	Login/OTP/session restore.	OTP/session/device indexes.	Authenticated skeletons.	Expired/wrong OTP, refresh rotation, role denial.	End-to-end login passes.
Phase 3 - Doctor Identity & Verification	Doctor profile/registration/qualification/experience/specialization/career/privacy + verification APIs.	Onboarding/profile/verification screens.	Doctor normalized schema.	Verified doctor E2E.	Duplicate registration, rejection/resubmit, privacy.	First milestone verified in API + app + admin.
Phase 4 - Hospital Domain & Admin Operations	Hospital/user/department/branch-ready model; verification; Django Admin workflows.	Mobile: hospital public profile. CRM: onboarding/profile/team.	Hospital tenant schema.	Verified hospital E2E.	Tenant isolation + verification tests.	Hospital can onboard and be approved.
Phase 5 - Network, Search & Privacy	Doctor search, connections, block/report baseline.	Mobile: network/search/profile. CRM: candidate search shell.	Connection/report indexes.	Professional network.	Block/privacy/duplicate request tests.	Network complete.
Phase 6 - Feed & Communities	Posts/media/comments/like/save/report; communities/membership.	Mobile: home/create/detail/comments/community. Admin: moderation.	Post/media/community tables.	Professional content loop.	Clinical confirmation/media/report tests.	Content MVP complete.
Phase 7 - Jobs Marketplace	Job CRUD/status/search; file CV support.	Mobile: jobs list/detail/save. CRM: jobs list/create/edit/detail.	Jobs/requirements relations.	Published jobs visible to doctors.	Ownership/status/range tests.	Job publish/browse works.
Phase 8 - Applications & Recruitment CRM	Apply, consent, history, interviews/offers, transitions.	Mobile: apply + applications timeline. CRM: candidates/detail/pipeline.	Application/history/interview/offer.	Apply-to-hire flow.	Transition/duplicate/tenant tests.	Full hiring workflow passes.
Phase 9 - Matching	Deterministic MatchService + cache invalidation strategy.	Mobile: recommended jobs/match display. CRM: recommended candidates.	Optional persisted score snapshot.	Explainable score.	Factor unit tests.	Same inputs yield stable score.
Phase 10 - Availability Exchange	Availability/slots, requirement, matching, shift lifecycle.	Mobile: calendar/status/shift requests. CRM: availability search/requirement/confirm.	Availability/workforce/history.	Core differentiator E2E.	Time overlap/race/state tests.	Requirement -> completed shift passes.
Phase 11 -	Conversation/	Mobile:	Messaging/	Persistent	Participant	Messages and

Messaging & Notifications	message; notification/event workers; FCM.	inbox/thread/notification/deep links. CRM: messages/alerts.	notification/device.	communication.	auth/push retry/read state.	deep links reliable.
Phase 12 - Subscription Entitlements	Plan/feature/subscription + entitlement middleware; no gateway until D1.	Mobile: plan display only if needed. CRM: settings/plan capability UI.	Commercial tables.	Backend-gated features.	Expiry/downgrade tests.	No hardcoded frontend entitlements.
Phase 13 - Security, Audit & Hardening	Audit coverage, rate limits, indexes, query optimization, backup/monitoring, deletion/export requests.	Mobile/CRM: offline, permission, error and accessibility polish.	Retention/index review.	Release candidate.	Security/performance/negative suite.	No critical/high release blockers.
Phase 14 - UAT & Production	Staging freeze, migrations, runbooks, production deploy.	Mobile closed beta + CRM design-partner release.	Backup restore rehearsal.	Production launch candidate.	UAT signoff + smoke + rollback.	Go-live checklist signed.

17. API Priority & Implementation Order

Priority	APIs
P0 - Blocking	Health/config; send/verify OTP; refresh/logout; files upload intent/finalize; doctor onboarding/profile/verification; hospital onboarding/profile/verification; admin verification decision; privacy baseline.
P1 - Core Business	Doctor search/connections; feed/posts/comments; communities; jobs CRUD/browse; apply/applications/transitions; availability; requirements/matches/shift lifecycle; report/block.
P2 - Supporting	Messaging; notifications; dashboard aggregates; saved jobs/posts; notification preferences; team management; subscription entitlements.
P3 - Enhancements	Advanced analytics; AI assistance; referrals/endorsements/events; advanced search engine; custom operations admin.

18. Frontend Development Sequence

Doctor Android Order	Hospital CRM Order
1 Splash/session -> Login -> OTP	1 Login/session -> Onboarding/verification
2 Doctor registration -> Verification status	2 CRM shell/sidebar -> Dashboard skeleton
3 Profile view/edit + privacy	3 Hospital profile -> Team
4 Network search -> Doctor profile -> Requests	4 Jobs list -> Create/Edit -> Detail
5 Home feed -> Create post -> Detail/comments	5 Candidates -> Candidate detail -> Pipeline
6 Communities list/detail/feed	6 Availability Exchange -> Requirement -> Matches -> Shift requests
7 Jobs list -> Detail -> Save/Apply -> Applications	7 Messages -> Notifications
8 Availability calendar -> Shift requests	8 Settings -> Subscription/entitlements
9 Messages -> Notifications/deep links	9 Error/permission/responsive polish
10 Settings/export/delete	10 UAT/admin support flows

19. Testing Strategy

- Backend unit: domain services, state transitions, permissions, matching factors, validators.
- API integration: every endpoint success + auth + validation + ownership + conflict + idempotency.
- Database: constraints, indexes, migrations forward/backward strategy, transaction race tests.
- Android: repository/use-case tests, critical UI flows, offline/retry, deep links, process death/session restore.

- CRM: forms, route guards, query states, tenant permissions, responsive critical pages.
- Security: OTP brute force, IDOR/object authorization, file upload bypass, sensitive field leakage, token replay/revocation, admin privilege tests.
- Performance: feed pagination, doctor/job search, candidate lists, requirement matching, N+1 checks.
- UAT: doctor verification; hospital verification; apply-to-hire; requirement-to-completed-shift; block/report; notification deep links.

Critical Scenario	Expected
Duplicate job apply	409; no duplicate application/history/notification
Two HR users transition same application	One valid transition; stale request gets 409; history consistent
Two hospitals request same doctor slot	Policy enforced; confirmation conflict handled atomically
Blocked doctor sends message	403/business violation
Private compensation via public doctor endpoint	Field absent, not null placeholder leakage
Invalid file disguised as PDF/image	Rejected before ACTIVE asset
FCM failure	Notification remains in DB unread; delivery retried/invalid token disabled
Hospital user accesses another hospital candidate	403/404 without data leakage

20. Production Architecture

Layer	Recommendation
Environments	Local -> Development -> Staging -> Production; separate credentials/databases/buckets.
Web/API	Django behind ALB/reverse proxy; EC2/ECS-style deployment acceptable initially; horizontal scaling later.
Database	AWS RDS PostgreSQL Multi-AZ when production criticality warrants; automated backups + PITR.
Cache/Queue	Managed Redis preferred; separate Celery workers/beat where needed.
Storage	Private S3 buckets; CloudFront for public/authorized media patterns where appropriate.
Observability	Structured logs, request IDs, metrics, uptime checks, error tracking, queue depth and DB health alerts.
CI/CD	GitHub Actions: lint/test/build/migration checks -> deploy dev/staging -> gated production.
Secrets	AWS Secrets Manager/SSM or equivalent; rotate; no .env secrets committed.
Backups	Automated DB backup + restore drill; S3 versioning/lifecycle as policy requires.
Deployment Safety	Pre-deploy migration review; backward-compatible changes; smoke tests; rollback/runbook.

21. Gap & Conflict Analysis

Type	Finding	Recommended Decision
Duplicate	All three docs repeat identity/network/jobs/availability/messaging/notifications/admin.	Consolidated into canonical domains and one development order.
Conflict	Generic PRD says healthcare professionals, while technical spec explicitly excludes students/nurses and says doctors only MVP.	MVP = registered doctors only; other professional types future.
Conflict	Hospital organization types vary (hospital/clinic/nursing home/diagnostic/medical college).	MVP UI supports Hospital/Clinic/Nursing Home; schema uses extensible type master for future categories.
Conflict	Admin roles: one MVP admin dashboard vs granular admin roles later.	Use Django Admin UI now but backend RBAC permissions from day one; granular UI groups can expand later.
Conflict	Professional Status appears like one field, but technical spec notes multiple simultaneous intents.	Canonical model = DoctorCareerPreference relation + derived status labels; not single enum.

Conflict	Messaging realtime approach unspecified/polling/Firebase possible.	MVP persistence is backend canonical; use polling/realtime adapter pragmatically; no WebSocket dependency.
Gap	Exact API schemas/responses/status codes were missing.	Defined versioned contract, standard errors and representative transactional APIs in this blueprint.
Gap	Exact DB columns/constraints/history tables incomplete.	Added normalized schema, unique constraints, status histories, consent and audit entities.
Gap	File size/type/security policy missing.	Added recommended file policy; max counts/retention remain configurable/Decision Required where source silent.
Gap	Online subscription payment workflow not specified.	Entitlement architecture only for MVP; payment gateway is Decision Required D1.
Gap	Hospital branches are needed for enterprise future but not consistently specified.	Add branch-ready model now; UI may hide branches for single-site MVP hospitals.
Gap	Cancellation/no-show rules for confirmed shifts unclear.	Keep lifecycle/audit now; cancellation penalties/reliability scoring are post-MVP until business policy approved.
Gap	Data retention/export/deletion operational rules not detailed.	Add request/consent/audit architecture; legal retention durations require compliance approval.
Gap	Email/SMS/WhatsApp fallback notification policy unclear.	FCM + in-app required; email/SMS/WhatsApp channels are configurable integrations after channel/legal decision.
Security Gap	Sensitive candidate/profile data could be exposed if filtering is frontend-only.	All visibility and tenant rules enforced in serializer/service/query layer; public serializers exclude sensitive fields.
Workflow Gap	Application/shift status could be overwritten without history.	Immutable ApplicationStatusHistory and ShiftStatusHistory required.
Frontend Gap	Source docs list modules but not loading/empty/error/success states.	Canonical screen tables require each page to implement all four states and action feedback.

22. Traceability Matrix

Requirement	Module	Frontend Screen	Fields	Backend API	Database	Role	Phase
OTP Login	Auth	Mobile Number/OTP; CRM Login	phone, otp	POST send-otp; verify-otp	users, otp_challenges, refresh_sessions	Doctor/Hospital User	P2
Doctor Verification	Doctor Identity	Registration/Verification	registration_number,council, year	doctor onboarding + verification submit	doctor_profiles, doctor_registrations, verification/audit	Doctor/Admin	P3
Professional Profile	Doctor Identity	Profile	headline,about,qualification,specialization,experience	GET/PATCH doctors/me	doctor_*	Doctor	P3
Connections	Network	Network/Doctor Profile	receiver_id	connection APIs	connections	Doctor	P5
Professional Feed	Content	Home/Create Post	category,body,clinical flag,media	feed/posts/comments	posts/post_media/comments/...	Doctor/Moderator	P6
Communities	Communities	Community List/Detail	community_id	community join/feed	communities, memberships	Doctor/Admin	P6
Hospital Profile	Hospital	CRM Onboarding/Profile; Mobile Hospital	name,type,location,beds,etc	hospital profile APIs	hospitals/departments/users	Hospital/Admin/Doctor view	P4

		Profile					
Jobs	Jobs	CRM Job Form; Mobile Jobs	job fields	hospital jobs; jobs browse/detail	jobs + requirement relations	Hospital/ Doctor	P7
Application	Applications	Job Detail/Applications; CRM Candidates	job_id,cv	apply/mine/transition	job_applications/history/interview/offer	Doctor/ Hospital	P8
Matching	Matching	Recommended Jobs/Candidates	profile/job factors	job detail/matches	read doctor/job/availability	Doctor/ Hospital	P9
Availability	Availability	Mobile Calendar; CRM Exchange	type,slots,radius,compensation	availability APIs	doctor_availability,slots	Doctor/ Hospital view	P10
Urgent Shift	Workforce	CRM Requirement/Shift; Mobile Shift Detail	requirement fields,doctor_id	requirements/matches/shift actions	urgent_requirements,shift_requests/history	Hospital/ Doctor	P10
Messaging	Messaging	Messages	conversation/body/file	conversation/message APIs	conversations/participants/messages	Doctor/ Hospital	P11
Notifications	Notifications	Notification Inbox/CRM alerts	read state	notifications APIs	notifications/device_tokens	All	P11
Privacy/Report	Governance	Profile Settings/Report	visibility/reason	privacy/report/block	privacy,consent,reports,blocks	All/Admin	P5/P13
Subscription	Commercial	CRM Settings	plan	subscription/entitlement	plans/features/subscriptions	Hospital/ Admin	P12

23. Final Master Development Checklist

- ☐ Architecture: modular monolith, module ownership, service boundaries, API versioning approved
- ☐ Database: ERD, UUIDs, constraints, indexes, state histories, migrations reviewed
- ☐ Backend: P0/P1 APIs implemented with OpenAPI, permissions, transactions and idempotency
- ☐ Mobile: auth -> identity -> network/feed -> jobs -> availability -> messaging/notifications complete
- ☐ CRM: onboarding -> profile -> jobs -> candidates -> availability -> messaging/team/settings complete
- ☐ Admin: doctor/hospital verification, moderation, reports, communities, restrictions and audit usable
- ☐ Files: S3 private/public policy, presigned flow, validation, signed download, cleanup workers
- ☐ Notifications: persistent table, FCM, deep links, read state, retry/token invalidation
- ☐ Security: rate limits, RBAC/IDOR tests, sensitive-data review, secret management, dependency scanning
- ☐ Testing: unit/API/DB/mobile/CRM/integration/security/performance/UAT gates passed
- ☐ Observability: logs, request IDs, errors, metrics, alerts and audit coverage live
- ☐ Deployment: dev/staging/prod separation, CI/CD, backups, restore test, migration/rollback runbook
- ☐ Legal/Policy: Terms, Privacy, Community Guidelines, Professional Conduct policy and retention decisions approved before public launch
- ☐ Decision Required items D1/D2 and any new unresolved business decisions closed or explicitly deferred

Appendix A - Canonical State Machines

Doctor/Hospital Verification:

UNVERIFIED -> PENDING -> VERIFIED

|-> REJECTED -> PENDING (resubmission if allowed)

Application:

APPLIED -> PROFILE_VIEWED -> SHORTLISTED -> INTERVIEW -> OFFERED -> HIRED

|-----> REJECTED

Doctor may -> WITHDRAWN from allowed pre-terminal states

Shift:

REQUESTED -> ACCEPTED_BY_DOCTOR -> CONFIRMED_BY_HOSPITAL -> COMPLETED

|-> DECLINED (recommended explicit state)

Any allowed active state -> CANCELLED

Assumption A3

The source flow names CANCELLED but not DECLINED as a stored status even though doctor has a Decline action. Recommended canonical implementation adds DECLINED as an explicit terminal state for analytics and audit.

Appendix B - Naming Conventions

Layer	Convention	Example
Frontend label	Human readable	Registration Number
API JSON	snake_case	registration_number
Python model field	snake_case	registration_number
Database column	snake_case	registration_number
Public identifier	UUID	doctor_id / job_id
Enums	UPPER_SNAKE_CASE	ACCEPTED_BY_DOCTOR
Timestamps	*_at in UTC	created_at, confirmed_at
Money	decimal + ISO currency	compensation + currency=INR

Appendix C - Source Reconciliation

The master blueprint is derived from the three supplied Doctors Network documents: the concept/business plan, the PRD, and the Product & Technical Specification. The technical specification is treated as the strongest implementation source when it narrows the broader business/PRD scope. Where all sources are silent, this blueprint uses labelled architecture recommendations, assumptions or Decision Required items rather than presenting invented business facts.